

Fisher, Markowitz, and Tobin

Three separation theorems, one underlying idea

EC3108 FED: Mainstream Static Supplementary Notes

One Story, Two narratives

Read across each row: the same four moves, applied first to time, then to risk.

	Step 1: frontier	Step 2: preference-free choice	Step 3: separating variable	Step 4: preference-dependent choice
Fisher (time)	Endowment + technology gives PPF $Y_1 = f(Y_0)$	Firm maximizes NPV at r , giving (Y_0^*, Y_1^*)	Wealth W_0 and a new budget line	Individual maximizes $U(C_0, C_1)$, giving lending S
Markowitz + Tobin (risk)	Risky assets give the Markowitz frontier	Bank rate r_f gives tangency portfolio w_T	(μ_T, σ_T) gives the Capital Market Line	Investor maximizes $V(\mu, \sigma)$, giving allocation x^*

Fisher, Step 1: The Environment

Two dates, 0 and 1. Endowed with $(Y_{0,\text{endow}}, Y_{1,\text{endow}})$. A technology lets you trade one for the other, with diminishing returns.

The production possibility frontier

$Y_1 = f(Y_0)$, with $f' > 0$ and $f'' < 0$ - more invested today buys more tomorrow, at a shrinking rate.

Notation: what is I ?

$I = Y_{0,\text{endow}} - Y_0$ - simply how much of today's endowment is handed to the production process rather than consumed immediately.

Fisher, Step 2: The Production Decision (Preference-Free)

Pick the point on the PPF that maximizes net present value at the market rate r - a decision needing no knowledge of anyone's tastes.

$$\max_{Y_0} -Y_{0,\text{endow}} + Y_0 + \frac{Y_1}{1+r}, \quad Y_1 = f(Y_0)$$

$$\text{F.O.C.:} \quad f'(Y_0^*) = 1 + r$$

Why this is the key step

(Y_0^*, Y_1^*) depends only on technology f and the market rate r - never on preferences. Everyone agrees on the same production plan.

Fisher, Step 3: Wealth- the Separating Variable

Collapse (Y_0^*, Y_1^*) into a single number: the present value of the optimal plan.

$$W_0 = Y_0^* + \frac{Y_1^*}{1+r}$$

New budget line: $C_1 = (1+r)(W_0 - C_0)$

A straight line through (Y_0^*, Y_1^*) with slope $-(1+r)$. W_0 is the one number the production stage hands forward — preferences have not entered yet.

Fisher, Step 4: The Consumption Decision (Preference-Dependent)

Only now do tastes enter.

$$\max_{C_0, C_1} U(C_0, C_1) \quad \text{s.t.} \quad C_0 + \frac{C_1}{1+r} = W_0$$

$$\mathcal{L} = U(C_0, C_1) + \lambda \left[W_0 - C_0 - \frac{C_1}{1+r} \right] \quad \Rightarrow \quad \frac{U_{C_0}}{U_{C_1}} = 1 + r$$

(C_0^*, C_1^*) depends on U ; savings $S = Y_0^* - C_0^*$. Different people land on different points of the *same* line.

The Fisher Separation Theorem

Theorem

The production decision (Y_0^*, Y_1^*) can be made independently of preferences - by maximizing wealth at r - before the consumption decision (C_0^*, C_1^*) is made by maximizing utility against the resulting wealth constraint.

A note on two agents

Why does a representative-agent model still keep “the firm” separate? Because the theorem separates two *decisions*, not two people. Even a lone agent who both produces and consumes solves it in two stages: first the production choice (needs no knowledge of tastes), then the consumption choice (needs only that). Calling the first stage “the firm” just makes the separation visible.

Why Markowitz Has to Come First

Tobin's theorem sounds like two steps: find the best risky combination, then mix it with a bank account. But "the best risky combination" is meaningless until a frontier of risky combinations already exists.

Year	What was established
1952	Markowitz shows how risky assets alone combine into an efficient frontier.
1958	Tobin adds a risk-free asset to that frontier and finds the tangency portfolio.

Exactly as Fisher's PPF must exist before the production decision can be discussed, Markowitz's frontier must exist before Tobin's tangency portfolio can be defined. So here, Markowitz *is* Step 1.

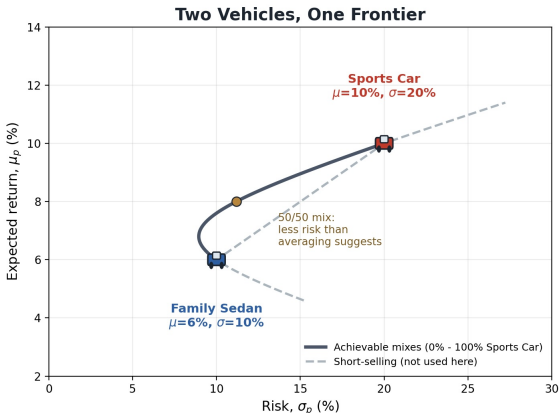
Step 1a: Two Risky Assets- A Frontier Is Born

Sports Car: $\mu_1 = 10\%$, $\sigma_1 = 20\%$. Family Sedan: $\mu_2 = 6\%$, $\sigma_2 = 10\%$. Uncorrelated.

$$\mu_p = w\mu_1 + (1 - w)\mu_2$$

$$\sigma_p^2 = w^2\sigma_1^2 + (1 - w)^2\sigma_2^2$$

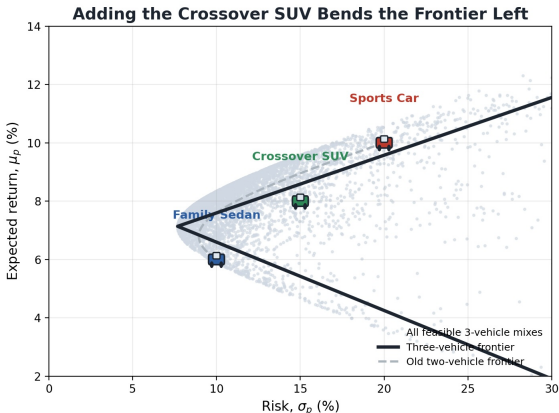
As w sweeps over all values, (σ_p, μ_p) traces a curve — the two-asset frontier.



Step 1b: Three Risky Assets - A Frontier of Frontiers

Add a Crossover SUV as a third asset ($\mu_3 = 8\%$, $\sigma_3 = 15\%$). Weights (w_1, w_2, w_3), $w_1 + w_2 + w_3 = 1$ — now two free parameters.

Every pair still traces its own curve; the true frontier is the upper-left boundary over *all* combinations — found by genuine constrained optimization, not by hand. Weights still sum to 1: fully invested in risky assets, no bank account yet.



Step 1c: The General n -Asset Problem

$$\min_w \frac{1}{2} w' \Sigma w \quad \text{s.t.} \quad w' \mu = \mu_p, \quad w' \iota = 1$$

$$\mathcal{L} = \frac{1}{2} w' \Sigma w - \lambda (w' \mu - \mu_p) - \delta (w' \iota - 1) \quad \implies \quad w^* = \lambda \Sigma^{-1} \mu + \delta \Sigma^{-1} \iota$$

Closed-form frontier

$$\sigma_p^2 = \frac{A\mu_p^2 - 2B\mu_p + C}{D}$$

where $A = \iota' \Sigma^{-1} \iota$, $B = \iota' \Sigma^{-1} \mu$, $C = \mu' \Sigma^{-1} \mu$, $D = AC - B^2$. This hyperbola, for any number of risky assets, is the Markowitz efficient frontier.

Without a riskfree bank, the direction Itself Rotates

$w^*(\mu_p) = \lambda(\mu_p) \Sigma^{-1} \mu + \delta(\mu_p) \Sigma^{-1} \iota$ — a blend of *two* fixed vectors, and *both* λ and δ move with the target μ_p . Watch what that does to the three-vehicle mix (Sports Car 10%, Family Sedan 6%, Crossover SUV 8%, uncorrelated):

Target μ_p	Sports Car	Family Sedan	Crossover SUV
7%	12.5%	62.5%	25.0%
8%	32.1%	32.1%	35.7%
9%	51.8%	1.8%	46.4%
10%	71.4%	-28.6%	57.1%

Family Sedan's share falls from 62.5% to almost nothing, then turns *negative* past $\mu_p \approx 9.06\%$: to hit a high enough target, you short the low-return vehicle and put the proceeds into the higher-return ones. The ratios among assets are genuinely *rotating*, not just scaling.

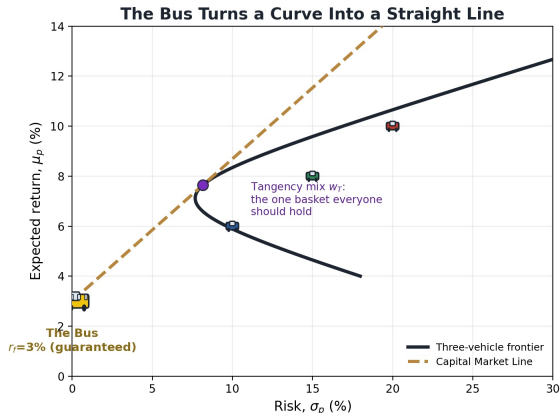
Compare to Tobin's $w^* = \theta \Sigma^{-1}(\mu - r_f \iota)$: one vector, one scalar θ — no component can ever flip sign relative to another. That contrast is exactly why adding the Bus is such a big deal.

Step 2: Enter the Bus - the Tangency Portfolio

The Bus offers $r_f = 3\%$ for certain, guaranteed arrival, no traffic drama and uncertainty! Mix a fraction x into *one* risky portfolio, $1 - x$ into Bus tickets.

$$w_T = \frac{\Sigma^{-1}(\mu - r_f \mathbf{1})}{\mathbf{1}' \Sigma^{-1}(\mu - r_f \mathbf{1})}$$

Found by maximizing the Sharpe ratio $(\mu_p - r_f)/\sigma_p$ subject to *one* constraint ($w' \mathbf{1} = 1$) - simpler than Markowitz, because the Bus absorbs the rest.



Why One Fixed Direction, for Every Target?

$$w^* = \theta \Sigma^{-1}(\mu - r_f \mathbf{1})$$

The vector $\Sigma^{-1}(\mu - r_f \mathbf{1})$ does not depend on θ - it is fixed by the data alone. Wanting more return never changes *which* assets you hold relative to each other; it only changes θ , how big a slice of wealth sits in that risky basket versus the Bus.

That is why every investor - cautious or aggressive - holds risky assets in exactly the same relative proportions (the tangency portfolio), differing only in how much of the whole pie is risky at all.

Contrast with pure Markowitz

Without a risk-free asset, $w(\mu_p)$ genuinely changes *direction* as the target changes - the ratios among assets can even flip sign.

Careful - Weights Don't All Sum to 1 Anymore

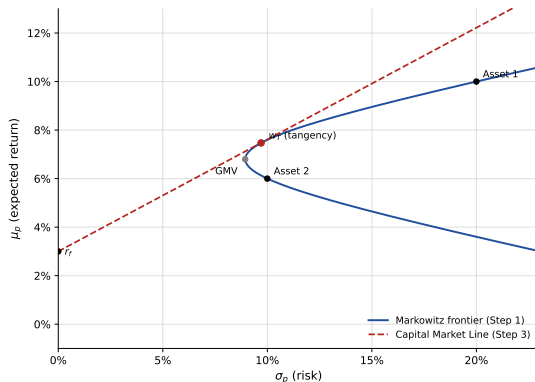
	Markowitz (no risk-free)	Tobin (with risk-free)
Constraint	$w_1 + \dots + w_n = 1$	Risky weights $w_{T1} + \dots + w_{Tn} = 1$ <i>among themselves</i>
Overall portfolio	Fully invested in risky assets only	$x \cdot w_T$ (risky) plus $(1 - x)$ in the Bus
Can $x > 1$?	Not applicable	Yes — borrow at r_f to lever up into the risky basket

The risky-only portion of the overall portfolio need not equal 1 — the Bus absorbs whatever is left over, whether lending ($x < 1$) or borrowing ($x > 1$).

Step 3: The Capital Market Line

With the tangency portfolio $T = (\sigma_T, \mu_T)$ pinned down, every efficient portfolio sits on one straight line running from the Bus through T .

$$\mu_p = r_f + \frac{\mu_T - r_f}{\sigma_T} \sigma_p$$



Step 4: Choosing x^* — the Preference-Dependent Step

Maximize mean-variance utility along the Capital Market Line.

$$\max_x V(x) = r_f + x(\mu_T - r_f) - \frac{\gamma}{2}x^2\sigma_T^2$$

Solution

$$x^* = \frac{\mu_T - r_f}{\gamma \sigma_T^2}$$

Higher risk aversion γ means smaller x^* — more wealth sits in the Bus. But everyone still holds the very same risky basket w_T ; only the split changes.

Two-Fund Separation — Back to Fisher

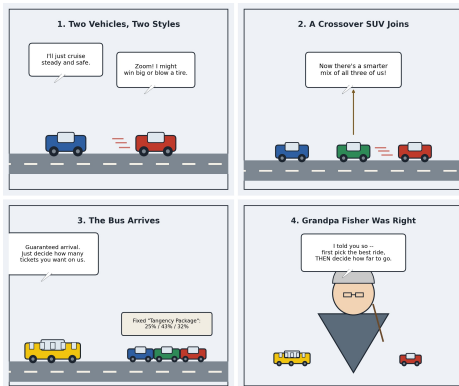
Two-Fund Separation Theorem

Every investor's optimal portfolio is built from just two funds: the risk-free asset and the single tangency portfolio T . Risk preferences (γ) determine only the split x^* between them — never the composition of the risky fund itself.

Fisher (time)	Role	Markowitz + Tobin (risk)
(Y_0^*, Y_1^*)	Preference-free choice	Tangency portfolio T
Wealth W_0 , budget line	Separating variable	Capital Market Line
(C_0^*, C_1^*) via U	Preference-dependent choice	x^* via γ

The Whole Story, Retold

Sports Car and Family Sedan build a frontier. Crossover SUV joins and the frontier gets better. The Bus arrives, and suddenly everyone should hold Sports Car, Family Sedan, and Crossover SUV in the very same ratio - only how many Bus tickets they buy differs. Grandpa Fisher was right all along!



- Fisher, I. (1930) *The Theory of Interest*. New York: Macmillan.
- Hirshleifer, J. (1958) 'On the Theory of Optimal Investment Decision', *Journal of Political Economy*, 66(4), pp. 329–352.
- Markowitz, H. (1952) 'Portfolio Selection', *Journal of Finance*, 7(1), pp. 77–91.
- Tobin, J. (1958) 'Liquidity Preference as Behavior Towards Risk', *Review of Economic Studies*, 25(2), pp. 65–86.